



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900

Purchases

18

Features

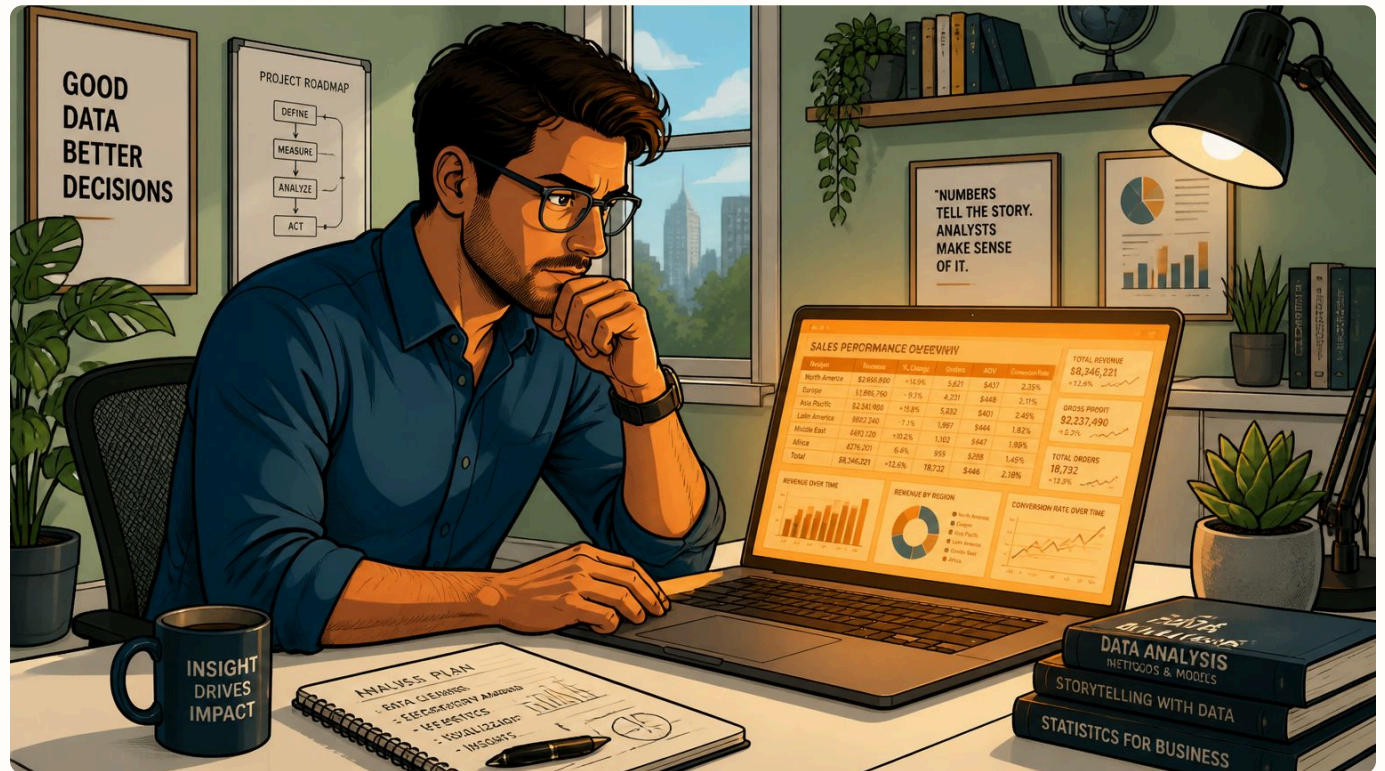
37

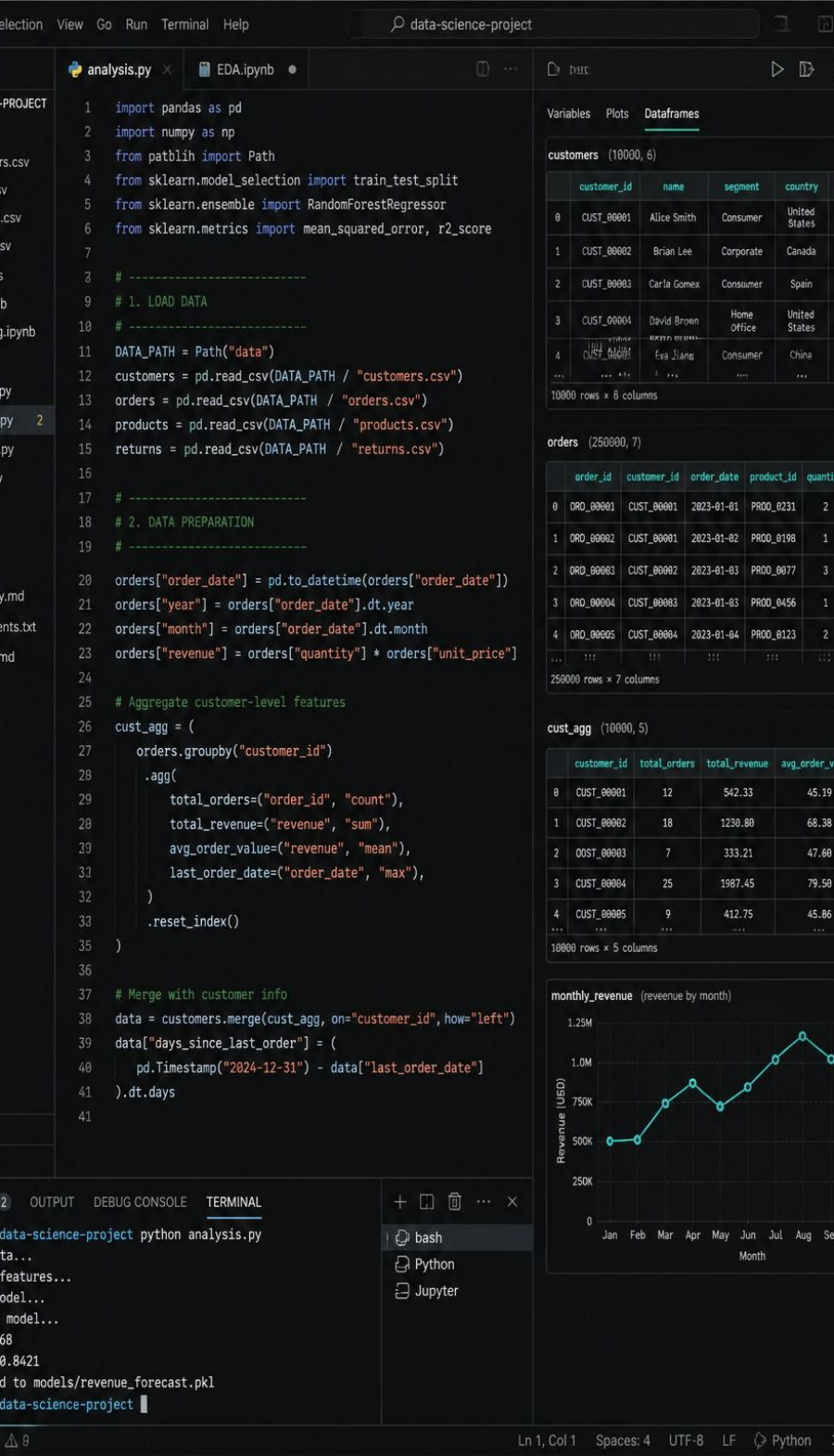
Missing Values

Review Rating only

Key Dimensions

- Demographics: Age, Gender, Location, Subscription
- Purchases: Item, Category, Amount, Season
- Behavior: Discount, Frequency, Review Rating





Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean

Impute Review Rating by category median; snake_case columns

03

Engineer

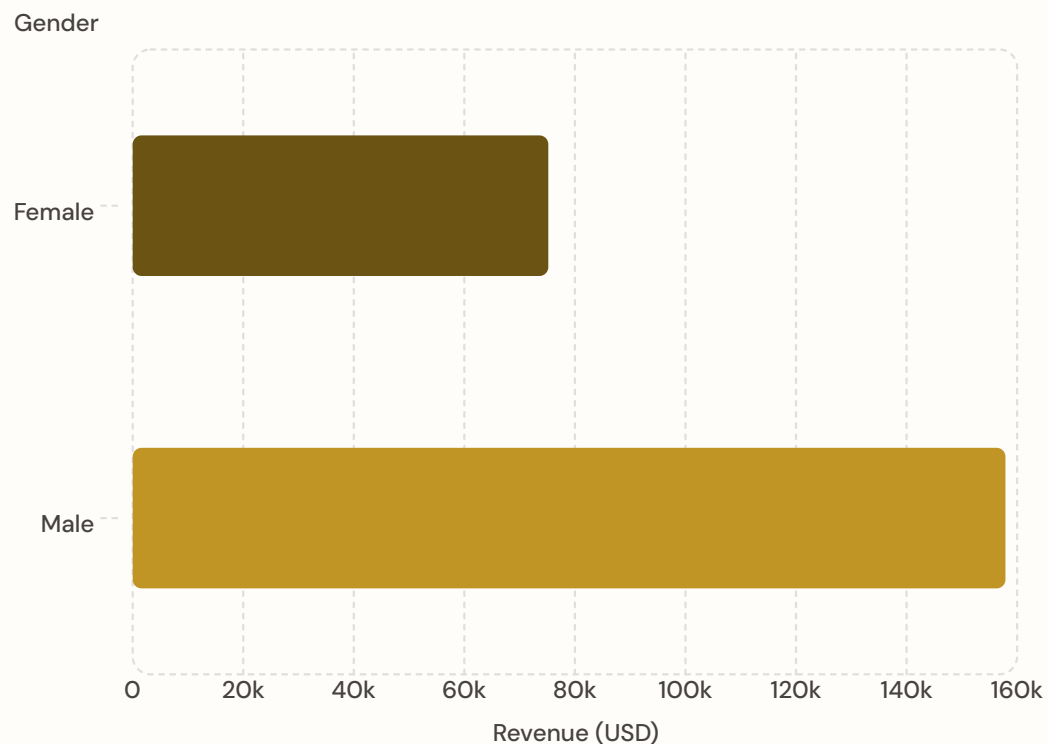
age_group bins, purchase_frequency_days

04

Integrate

Load cleaned DataFrame to PostgreSQL

Revenue by Gender



Male customers drive **2× more revenue** than female customers

Male: **\$157,890** · Female: **\$75,191**

Top Products & Discount Patterns

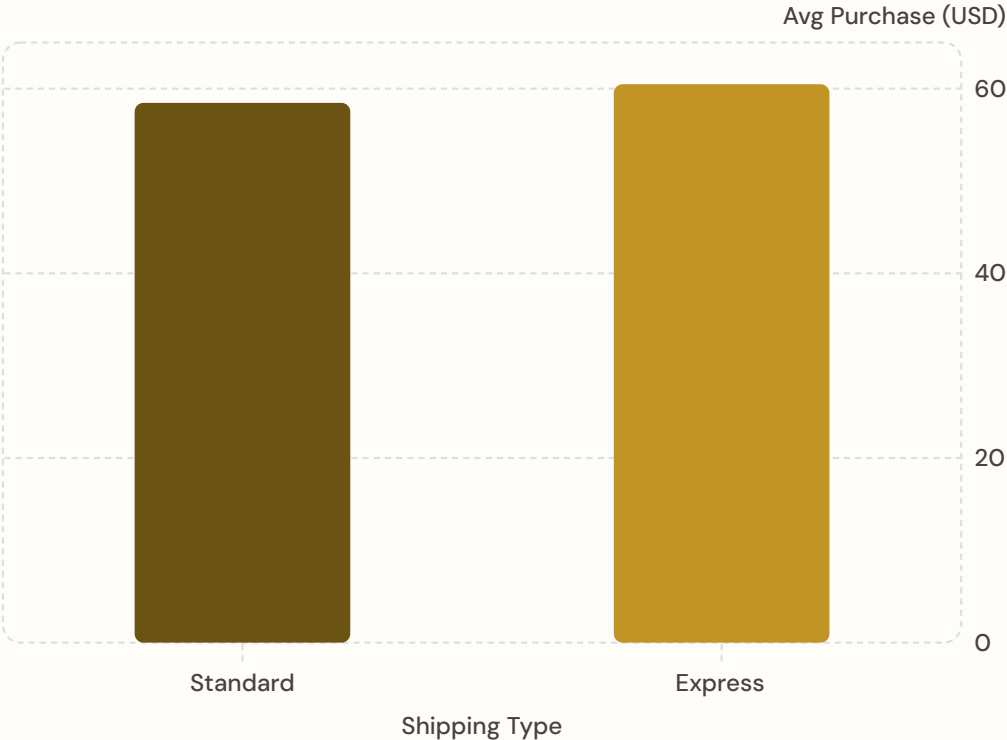
Top 5 by Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Item	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights



Subscribers vs. Non-Subscribers

Status	Customers	Total Revenue
Yes	1,053	\$62,645
No	2,847	\$170,436

Non-subscribers: **73%** of customers, **\$170K** revenue

Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

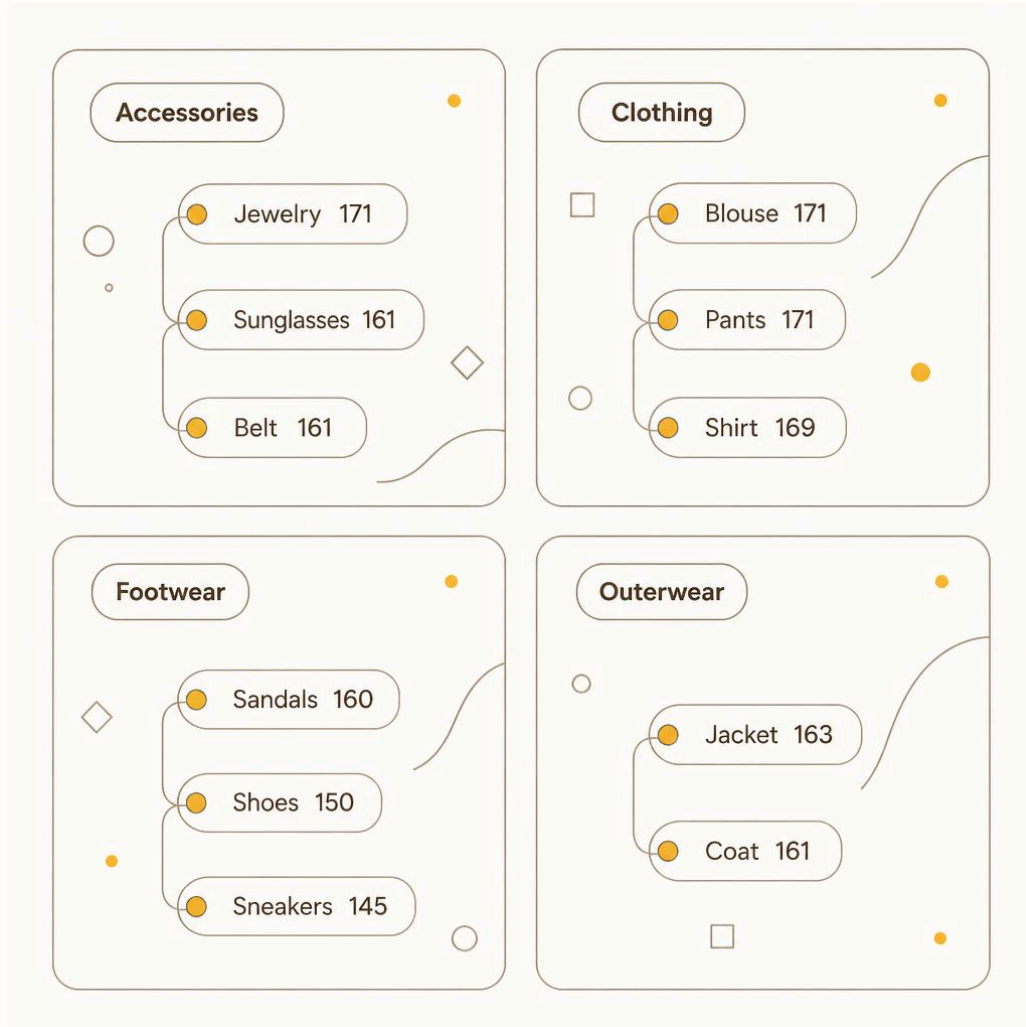
New

83 customers

Repeat buyers (>5 purchases): **958** subscribed vs. **2,518** not subscribed



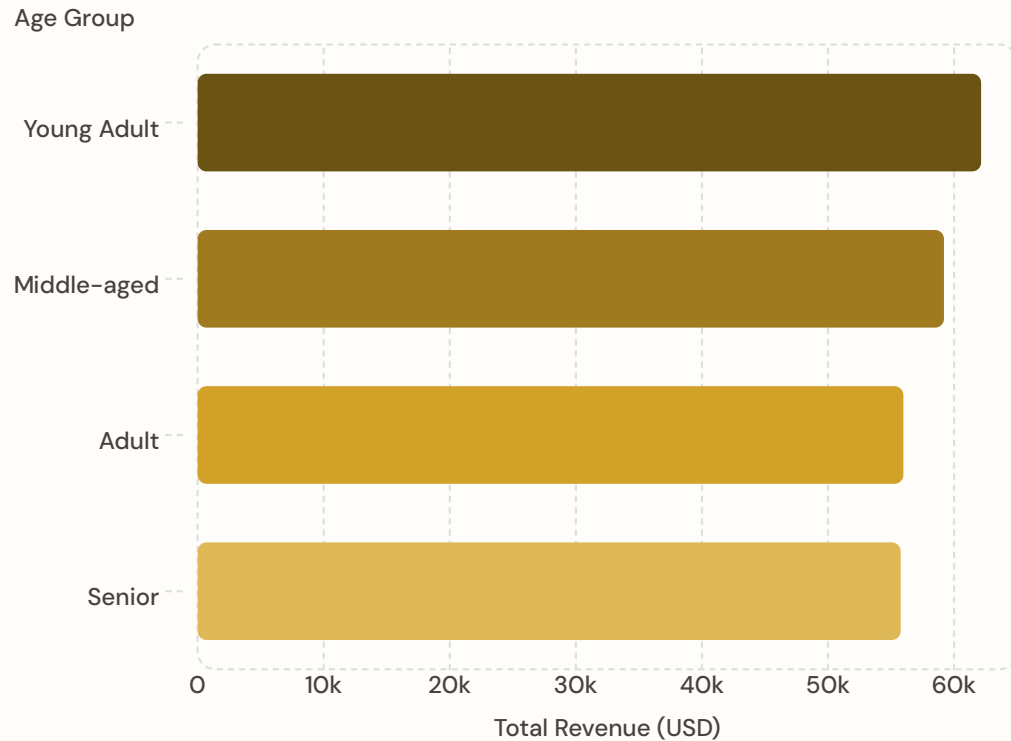
Top Products by Category



Category Leaders

- **Clothing** — Blouse, Pants, Shirt
- **Outerwear** — Jacket, Coat
- **Footwear** — Sandals, Shoes
- **Accessories** — Jewelry, Sunglasses

Revenue by Age Group



Key Takeaway

Young Adults lead revenue at **\$62,143**. Revenue declines gradually with age — a relatively even spread across groups.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

